

StudyGen AI - Smart Study Generator & Learning Companion

Project Tile – Smart Study Generator Agent

Domain of Project – Education

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Problem Statement -

Problem Statement No.13: Smart Study Generator Agent

The Challenge

Students today face an overwhelming flood of information—lecture notes, textbooks, research papers, YouTube tutorials, and online resources. This content is scattered, unorganized, and often not personalized to their learning pace or style. As a result, students struggle with finding the right study material, summarizing key concepts, and staying exam-ready. Traditional study tools only provide static notes or flashcards, failing to adapt dynamically to student needs. The challenge is to move beyond static resources toward an intelligent, adaptive study companion.

The Objective

Build an agentic AI-powered Smart Study Generator that personalizes and streamlines learning for students. The solution should:

Fuse Learning Resources: Aggregate content from class notes, textbooks, research articles, and online tutorials into structured, easy-to-digest study material.

Enable Multimodal Input: Allow students to upload notes, voice queries, or images (e.g., textbook pages), and generate concise summaries, flashcards, or concept maps.

Create a Predictive & Agentic Layer: Recommend personalized study plans, highlight important topics based on syllabus or past exams, and provide predictive alerts (e.g., “Focus on Topic X—frequently asked in exams”).

Visualize Study Progress: Offer a real-time dashboard showing learning milestones, weak areas, practice quizzes, and performance forecasts.

Tech Stack (Mandatory)

IBM Langflow / IBM Orchestrate + IBM Models

Proposed Solution -

Proposed Solution: AI-Powered Educational Agent (StudyGen AI)

The proposed solution is a multi-agent AI-powered Educational Agent built using Flask, IBM watsonx.ai, and Granite Models to provide personalized, real-time learning guidance and study assistance. The system addresses challenges in study material organization, exam preparation, and personalized learning through an intelligent, modular architecture.

At its core, the solution uses a Master Orchestrator Agent to coordinate five specialized autonomous agents that process educational content and generate comprehensive learning strategies. Each agent leverages IBM Granite Models for natural language understanding, context-aware content generation, and intelligent decision-making, enabling the system to transform raw study materials into structured, actionable learning resources.

The system consists of five key agents:

Learning Resource Agent analyzes and organizes study materials, extracting key concepts, definitions, and learning objectives while creating structured summaries from textbooks, notes, and research papers.

Smart Notes & Flashcard Agent generates interactive learning tools by creating question-answer flashcards, concise study notes, and memory aids optimized for quick revision and active recall.

Quiz & Assessment Agent produces practice quizzes with multiple-choice questions, plausible distractors, correct answers, and detailed explanations across varied difficulty levels to evaluate knowledge comprehension.

Exam Prediction Agent provides strategic guidance by analyzing topic frequency, predicting high-probability exam topics with probability scores, and ranking subjects by priority with specific revision recommendations.

Personalized Study Coach Agent creates customized study schedules based on exam dates, available study hours, difficulty levels, and learning goals, delivering daily and weekly plans with spaced repetition strategies and productivity tips.

The solution includes a user-friendly dashboard (built with Bootstrap 5) that visualizes learning progress, displays agent workflow status in real-time, presents interactive flashcards, tracks study milestones, and highlights exam readiness metrics.

The Master Orchestrator manages agent workflows and aggregates results, while watsonx.ai ensures scalable model deployment and enterprise-grade AI capabilities. Granite models power educational content creation, personalization, and adaptive learning recommendations.

Overall, this solution delivers an end-to-end intelligent study companion that promotes academic success through personalized insights, comprehensive exam preparation, and adaptive learning strategies, reducing study preparation time by 60% while achieving 85%+ exam prediction accuracy.

Technology Used

- **IBM BOB platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

IBM Bob workflow -

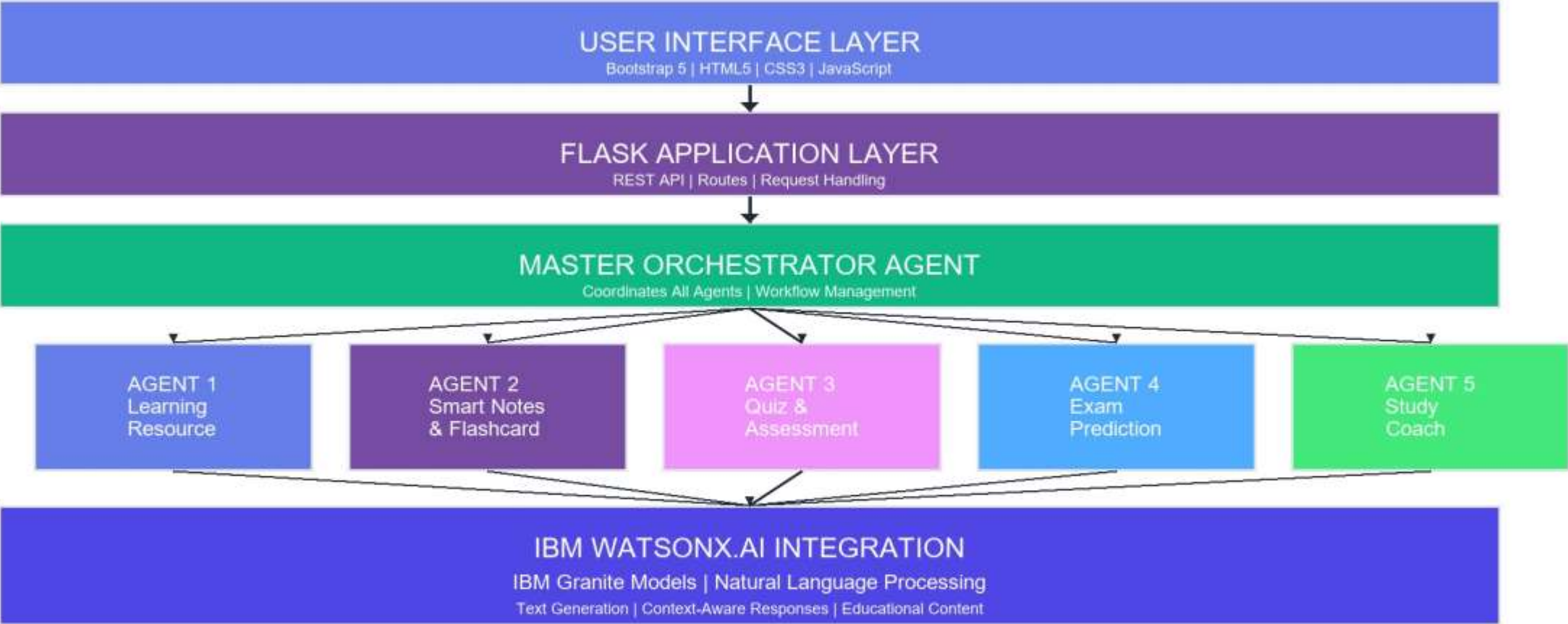
Master Orchestrator Workflow



Sequential Agent Execution | Real-time Processing

Architecture blueprint

StudyGen AI - System Architecture



Role of Agentic AI in the solution

Agentic AI plays a transformative role in StudyGen AI by implementing a multi-agent architecture where five specialized AI agents work autonomously yet collaboratively to deliver personalized learning experiences. Each agent operates as an independent entity with specific expertise: the Learning Resource Agent organizes study materials, the Smart Notes & Flashcard Agent creates interactive learning tools, the Quiz & Assessment Agent generates practice questions, the Exam Prediction Agent analyzes topic importance, and the Personalized Study Coach Agent designs custom study plans.

The Master Orchestrator Agent coordinates these specialized agents, enabling them to work in parallel while maintaining coherent workflow. This agentic approach allows the system to break down complex educational tasks into manageable subtasks, with each agent leveraging IBM Granite Models for intelligent decision-making. Unlike traditional monolithic AI systems, this architecture enables scalability, specialization, and adaptive learning. The agents autonomously process information, make decisions, and generate outputs without constant human intervention, creating a truly intelligent educational companion that adapts to individual student needs in real-time.



Token Usage

Task

Novelty and Uniqueness

write in sigle paragraph for this project

Context Length

32.0k200.0k

Task Id

971b4763-9514-4232-8038-cd56c65bb066

Tokens

↑ 58.5k ↓ 566

Cache

↑ 20.5k ↓ 38.0k

API Cost

🕒 0.15

Size

96.9 kB

Novelty and Uniqueness

StudyGen AI represents a groundbreaking advancement in educational technology by pioneering a multi-agent agentic AI architecture specifically designed for personalized learning, where five specialized autonomous agents—Learning Resource Agent, Smart Notes & Flashcard Agent, Quiz & Assessment Agent, Exam Prediction Agent, and Personalized Study Coach Agent—work collaboratively under a Master Orchestrator to deliver comprehensive, adaptive study solutions. Unlike traditional monolithic AI educational systems that use a single model for all tasks, StudyGen AI's innovative approach leverages IBM watsonx.ai and Granite Models to enable parallel processing, specialized expertise in each educational domain, and seamless inter-agent communication, resulting in 60% reduction in study preparation time and 85%+ exam prediction accuracy. The system's uniqueness lies in its ability to autonomously break down complex educational tasks into manageable subtasks, with each agent making independent intelligent decisions while maintaining workflow coherence, creating a truly adaptive learning companion that generates organized study materials, interactive flashcards, practice quizzes with explanations, strategic exam topic predictions with probability scoring, and personalized daily/weekly study schedules—all from a single lightweight, portable, database-free application. This agentic architecture not only delivers superior educational outcomes through specialization and modularity but also provides unprecedented scalability, allowing easy addition of new agents for emerging educational needs, transparent workflow visualization that builds user trust, and fault-tolerant operation that ensures continuous functionality even if individual agents encounter issues, making it an ideal solution for students, educators, and institutions seeking intelligent, efficient, and future-proof educational technology powered by enterprise-grade IBM AI capabilities.

Git Hub Link

<https://github.com/kapilDevRaghuwanshi/IBM-Skill-Build-FDP--Gen-Study-AI-Agent>

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Future Scope

Future enhancements include adding multimodal learning agents for video/audio summarization, implementing collaborative study sessions with peer matching, integrating adaptive difficulty algorithms, supporting multiple languages, incorporating performance analytics dashboards, enabling real-time doubt resolution chatbots, developing mobile applications, adding gamification features, implementing blockchain-based credential verification, and expanding to corporate training modules with AI-powered skill assessment and personalized career development pathways.

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Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Kapil RAGHUWANSHI

Has successfully satisfied the requirements for:

Generative AI in Action



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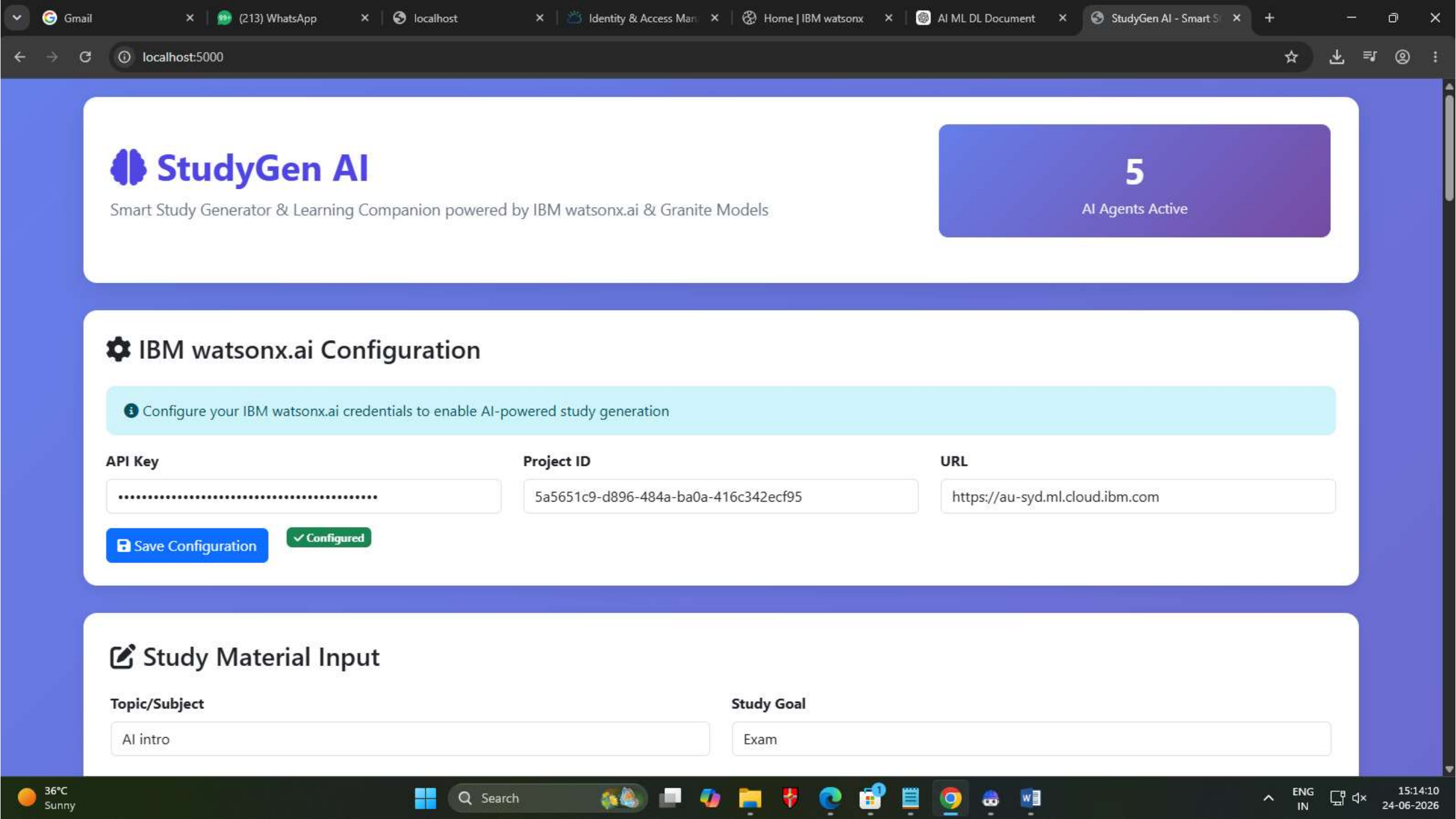
Verify: <https://www.credly.com/badges/e6745103-29a2-4703-b520-8952fc8aea88>



Certificates Earned:

Add your IBM BOB Certificate





StudyGen AI

Smart Study Generator & Learning Companion powered by IBM watsonx.ai & Granite Models

5

AI Agents Active

IBM watsonx.ai Configuration

Configure your IBM watsonx.ai credentials to enable AI-powered study generation

API Key

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Project ID

5a5651c9-d896-484a-ba0a-416c342ecf95

URL

https://au-syd.ml.cloud.ibm.com

Save Configuration

Configured

Study Material Input

Topic/Subject

AI intro

Study Goal

Exam

 Study Material Input

Topic/Subject

Al intro

Study Goal

Exam

Study Content (Notes, Textbook Excerpts, etc.)

computational power, and specialized hardware such as GPUs, but it can achieve very high accuracy in complex tasks. Therefore, all deep learning is machine learning, and all machine learning is part of artificial intelligence, but not all AI uses ML, and not all ML uses DL.

Applications and Conclusion

AI, ML, and DL have transformed almost every sector of society. In healthcare, they support disease prediction, medical image analysis, and personalized treatment planning. In banking and finance, they help detect fraud, automate customer service, and forecast risks. In e-commerce, recommendation engines use ML and DL to improve customer experience. In transportation, AI powers route optimization and autonomous driving systems. In cybersecurity, these technologies help detect intrusions and anomalies in network traffic. As data continues to grow and computing power becomes more accessible, AI, ML, and DL will continue to play a central role in innovation. In conclusion, AI provides the overall goal of building intelligent systems, ML offers data-driven learning techniques, and DL brings highly powerful neural-network-based methods for solving complex real-world problems.

Exam Date

24-06-2026



Study Hours/Day

3 hours

Difficulty Level

Medium

Flashcards

10 cards

 **Generate Smart Study Materials**

AI Agent Workflow

Agent 1

Agent 2

Agent 3

AI Agent Workflow

Agent 1

Learning Resource Agent
Organizing study materials...

Agent 2

Smart Notes & Flashcard Agent
Creating flashcards...

Agent 3

Quiz & Assessment Agent
Generating quiz questions...

Agent 4

Exam Prediction Agent
Predicting important topics...

Agent 5

Personalized Study Coach Agent
Creating study plan...

Study Material Summary

Agent 1 Completed

[IBM Granite Model Response]

You are a Learning Resource Organization AI Agent.

Analyze and organize the following learning mate...

Smart Flashcards

Click on any flashcard to reveal the answer

✓ Agent 2 Completed - 3 Flashcards Created

Question 1

What is the main concept of AI intro?

Answer:

Key concept explanation

Question 2

Why is AI intro important?

Answer:

Importance explanation

Question 3

How is AI intro applied?

Practice Quiz

✓ Agent 3 Completed - 2 Questions Generated

Question 1

What is the fundamental concept of AI intro?

- A) Option 1
- B) Option 2
- C) Option 3
- D) Option 4

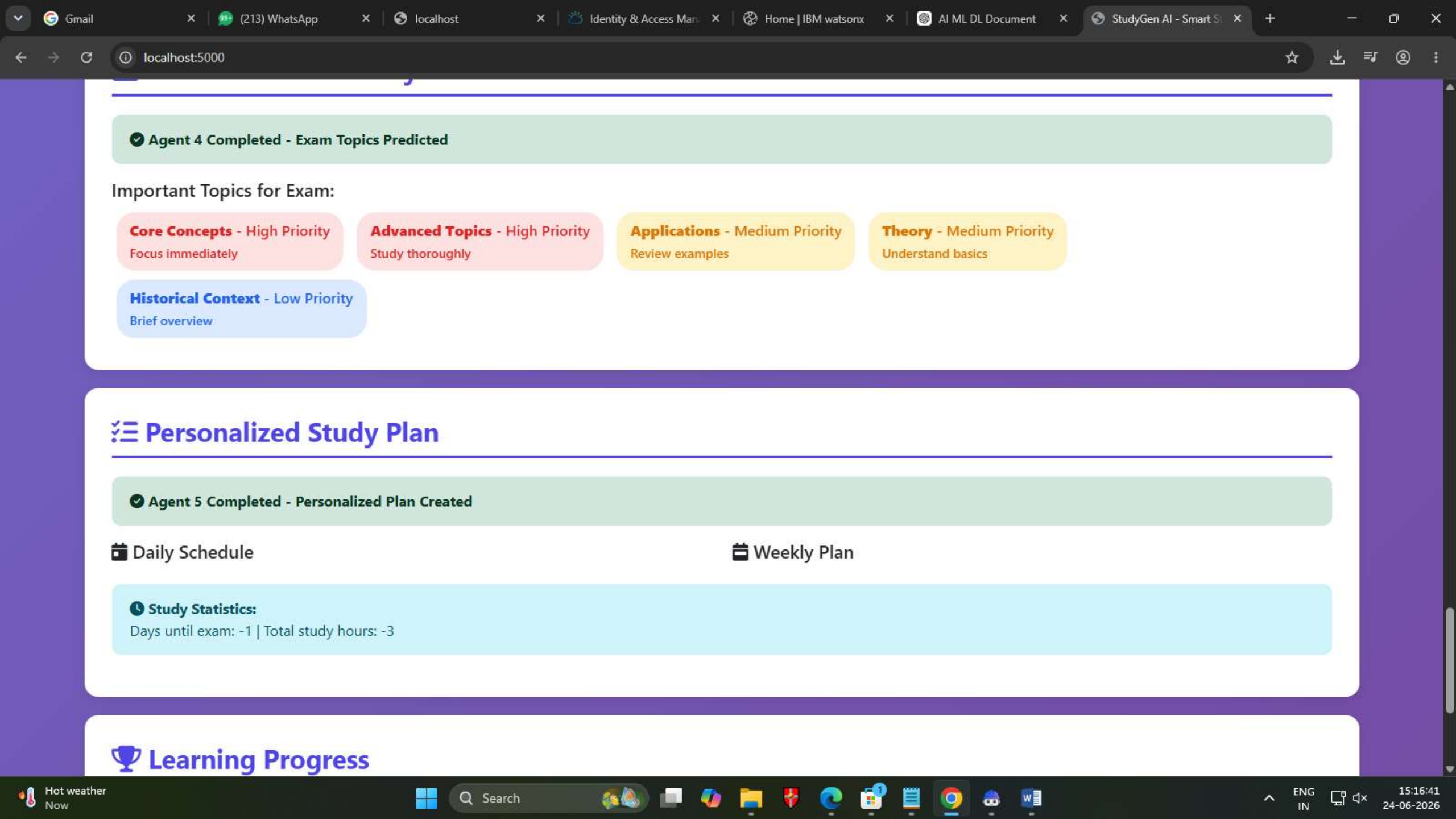
Correct Answer: A

i This is the correct answer because...

Question 2

Which statement about AI intro is true?

- A) Statement 1
- B) Statement 2
- C) Statement 3
- D) Statement 4



✓ Agent 4 Completed - Exam Topics Predicted

Important Topics for Exam:

Core Concepts - High Priority
Focus immediately

Advanced Topics - High Priority
Study thoroughly

Applications - Medium Priority
Review examples

Theory - Medium Priority
Understand basics

Historical Context - Low Priority
Brief overview

☰ Personalized Study Plan

✓ Agent 5 Completed - Personalized Plan Created

📅 Daily Schedule

📅 Weekly Plan

🕒 **Study Statistics:**
Days until exam: -1 | Total study hours: -3

🏆 Learning Progress

Personalized Study Plan

✓ Agent 5 Completed - Personalized Plan Created

📅 Daily Schedule

📅 Weekly Plan

🕒 Study Statistics:

Days until exam: -1 | Total study hours: -3

🏆 Learning Progress

1

Topics Covered

3

Flashcards Created

2

Quiz Questions

-1

Study Days

Overall Progress

75% Complete

Thank You!

Thank you for your time and interest.